



TEMASEK JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATION
Higher 1

CHEMISTRY

8873/01

15 September 2021
1 hour

Additional Materials: Multiple Choice Answer Sheet (OMS)
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Class and index number on the Answer Sheet in the spaces provided.

1. Enter your NAME (as in NRIC). _____
2. Enter the SUBJECT TITLE. _____
3. Enter the TEST NAME. _____
4. Enter the CLASS. _____

Write your **name**
and **Civics Group**

Write and shade
your index number

WRITE		SHADE APPROPRIATE BOXES									
I N D E X		0	1	2	3	4	5	6	7	8	9
		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
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		A	B	C	D	E	F	G	H	I	
		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **12** printed pages.

- What is the change in oxidation number of sulfur in this reaction?

B +4 to +6

D +6 to +4

- A** 1.75×10^{23}

B 3.01×10^{23}

C 3.50×10^{23}

D 6.02×10^{23}

- $${}_{88}^{223}\text{Ra} \longrightarrow \text{Q} + {}_2^4\text{He}$$

A 219

B 135

C 133

D 86

- 1** This shell can accommodate a maximum of eight electrons.

2 It contains three subshells and nine orbitals.

3 All the orbitals are directional and degenerate.

A 2 only

B 1 and 2 only

C 1 and 3 only

D 1, 2 and 3

- Which particle would, on gaining an electron, have a fully-filled set of p orbitals?

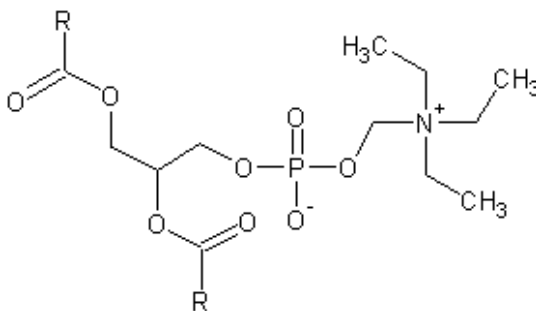
A F^+

B **O⁻**

C S^+

D P⁻

- 6** Lecithin is an emulsifier added to chocolate to help bind the cocoa solids, sugar and milk so they stick to the cocoa butter. Its structure is shown below.

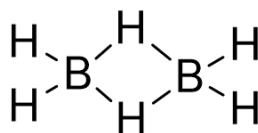


R: long hydrophobic carbon chains

Which of the following statements about lecithin is correct?

- A** It can form instantaneous dipole-induced dipole interactions with non-polar molecules present in chocolate.
- B** It can form hydrogen bonds between its own molecules.
- C** All bond angles surrounding each carbon atom are 109.5° .
- D** It is square planar about P.

- 7 Diborane, B_2H_6 , is a colorless gas with a sweet odour. The structure is unique as it comprises of two bridging hydrogens shared between two boron atoms. The B_2H_2 ring contains two B-H-B bonds. Each B-H-B bond is held together by two electrons.



Diborane

Which of the following statements are correct about diborane?

- 1 It has 16 bonding electrons.
- 2 Only boron is electron deficient.
- 3 All B-H bonds in diborane have the same bond strength.

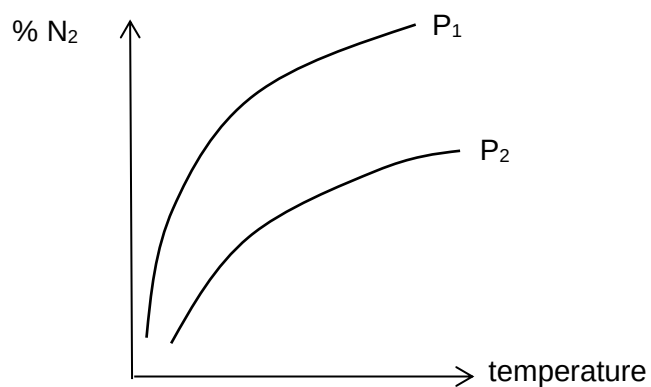
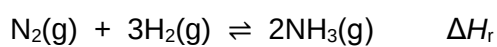
- A** 1 and 3 only **B** 1 and 2 only
C 2 and 3 only **D** 2 only

- 8 Which of the following compounds are arranged according to increasing boiling point?
- A NH_3 , H_2O , HF
- B H_2O , H_2S , H_2Se
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CH}_3\text{CH}_2\text{COOH}$
- D $(\text{CH}_3)_3\text{CCl}$, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{Cl}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$
- 9 A Period 3 chloride, **W**, reacts with water in two separate ways.
- When a few drops of water are added to **W**, steamy white fumes are evolved and a white solid remains, which is insoluble in water.
- When a large amount of water is added to **W**, a clear solution which turns Universal Indicator orange is obtained.
- Which compound is **W**?
- A MgCl_2 B AlCl_3 C SiCl_4 D PCl_5
- 10 **X** and **Y** are two elements in Period 3 of the Periodic Table. They combine to form compound **Z**. Both **X** and **Y** form oxides that react with aqueous sodium hydroxide. The oxidation number of **X** in its oxide is +4.
- What is the formula of compound **Z**?
- A AlP B Al_2S_3 C Si_3P_4 D SiS_2
- 11 The solids, sodium chloride and sodium iodide, both react with concentrated sulfuric acid at room temperature.
- With NaCl , the products are NaHSO_4 and HCl .
- With NaI , the products are NaHSO_4 , HI , I_2 , SO_2 , H_2O , S and H_2S .
- Which of the following is the best explanation for the difference in products?
- A Chloride will displace iodine from solution.
- B Iodine is a better oxidising agent than chlorine.
- C Iodide is a better reducing agent than chloride.
- D Sulfuric acid is able to act as a dehydrating agent with NaI .

12 Which of the following shows the trends of Group 1 elements, lithium to caesium?

- A** First ionization energy increases
- B** Electronegativity increases
- C** Reducing power increases
- D** Melting point increases

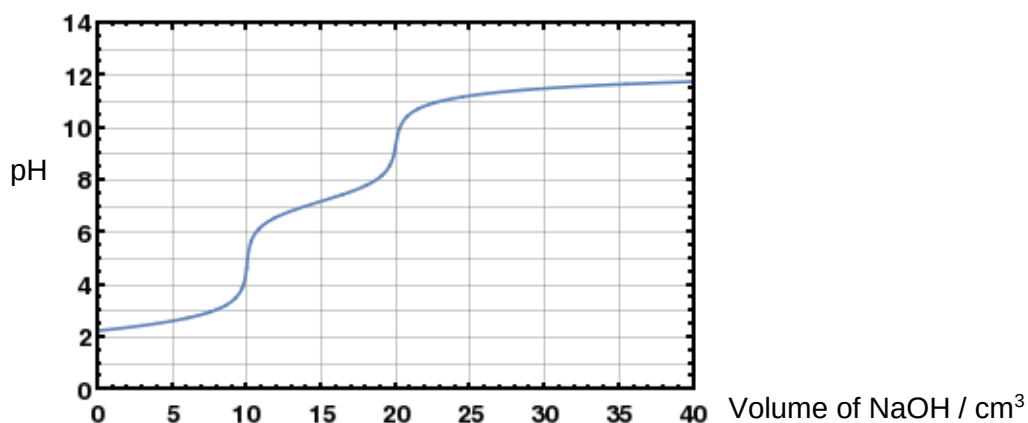
13 The graph below shows how the percentage of $\text{N}_2(\text{g})$ in the equilibrium mixture of the Haber process varies with temperature at different pressures, P_1 and P_2 .



Which row in the table is correct?

	ΔH_r	Pressure
A	positive	$P_1 < P_2$
B	positive	$P_1 > P_2$
C	negative	$P_1 < P_2$
D	negative	$P_1 > P_2$

- 14 When an acid is titrated with a strong base, an equivalence point is reached when the amount of base added is equal to the amount of H^+ dissociated. The following pH curve is obtained when 10.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ of a diprotic acid, H_2X , is titrated against $0.100 \text{ mol dm}^{-3}$ NaOH .



The table shows the pH range of colour changes of some acid-base indicators.

Indicator	pH range of colour changes
thymol blue	1.2 – 2.8 and 8.0 – 9.6
bromocresol green	3.8 – 5.4
phenol red	6.8 – 8.4

Which indicators should be used to show the first and second equivalence points?

- A thymol blue for both
- B thymol blue and then phenol red
- C bromocresol green and then phenol red
- D bromocresol green and then thymol blue

- 15 The position of equilibrium lies to the right in each of these reactions.



Based on this information, which statements are correct?

- 1 ClO^- is the conjugate base of HClO .
- 2 N_2H_5^+ is the Bronsted base in Reaction 2.
- 3 The order of acid strength is $\text{HClO} > \text{N}_2\text{H}_5^+ > \text{NH}_4^+$.
- 4 N_2H_4 is a stronger base than NH_3 .

- A 1 and 3 only
- B 1 and 4 only
- C 2 and 3 only
- D 2 and 4 only

- 16** A solution was made by mixing 0.002 mol of $\text{H}_2\text{SO}_4(\text{aq})$ and 0.005 mol of $\text{KOH}(\text{aq})$. Water was added until the volume of the solution was 1 dm^3 .

What is the pH of the solution?

- A** 11.0 **B** 11.5 **C** 11.7 **D** 12.0

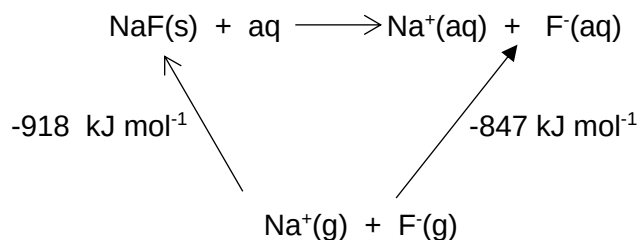
- 17** The lattice energies of rubidium fluoride, RbF , and caesium chloride, CsCl , are -760 kJ mol^{-1} and -650 kJ mol^{-1} , respectively.

Which value is likely to be the lattice energy of caesium fluoride, CsF , in kJ mol^{-1} ?

- A** -460 **B** -550 **C** -680 **D** -800

- 18** A student dissolved 8.4 g of sodium fluoride in 250 g of water.

The energy cycle for the dissolution is given below.



What would be the initial temperature of the water if the final temperature of the solution is 20.00°C ? Assume that the specific heat capacity of sodium fluoride solution is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

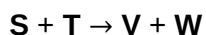
- A** 6.48°C **B** 20.01°C **C** 33.08°C **D** 33.52°C

- 19** In an experiment, the rate of the decomposition of hydrogen peroxide was investigated by collecting the oxygen gas produced. It was found that 10% of 0.20 mol dm^{-3} of H_2O_2 decomposed in 8 minutes at a particular temperature.

What is percentage of H_2O_2 decomposed in 8 minutes if 0.10 mol dm^{-3} of H_2O_2 was used?

- A** 5% **B** 10% **C** 20% **D** 50%

- 20** Consider the following reaction involving aqueous reactants **S** and **T**, and catalysed by **U**:



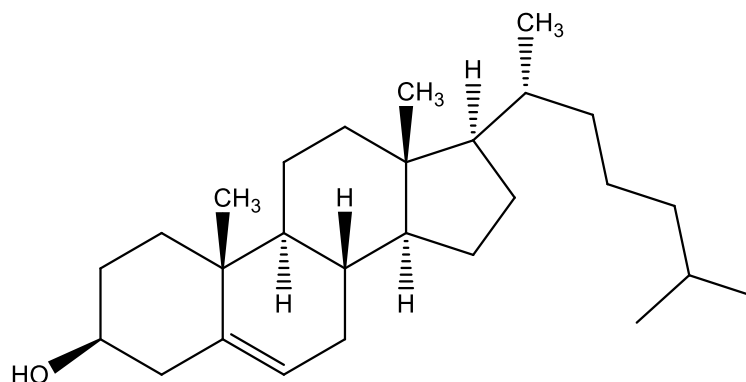
The solution is originally coloured due to reactant **T**. The kinetics of the above reaction is studied by measuring the time taken for the solution to decolourise. The results are obtained below.

Experiment	Volume of S / cm ³	Volume of T / cm ³	Volume of U / cm ³	Volume of water added / cm ³	Time taken / s
1	10	5	5	5	20
2	5	5	5	10	40
3	10	5	2.5	7.5	40
4	10	2.5	5	7.5	10

What is the rate law for the reaction above?

- A** Rate = $k[\mathbf{S}][\mathbf{T}][\mathbf{U}]$
- B** Rate = $k[\mathbf{S}][\mathbf{T}]$
- C** Rate = $k[\mathbf{T}][\mathbf{U}]$
- D** Rate = $k[\mathbf{S}][\mathbf{U}]$
- 21** Which of the following is true about the value of rate constant of a pseudo first order reaction?
- A** It depends on the concentration of reactants present in limited amount.
- B** It depends on the concentration of reactants present in excess amount.
- C** It is independent of the concentration of reactants.
- D** It depends only on temperature and presence of catalyst.

- 22 Cholesterol is a waxy substance that is present in blood plasma and in all animal tissues.



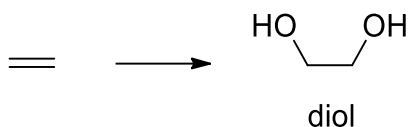
Cholesterol

Which of the following statements about cholesterol are **incorrect**?

- 1 All the seventeen carbon atoms in the four rings lie in the same plane.
- 2 It can exhibit cis-trans isomerism at the C=C double bond.
- 3 Its molecular formula is $C_{27}H_{46}O$.

- | | |
|-----------------------|-----------------------|
| A 1 only | B 1 and 2 only |
| C 2 and 3 only | D 1, 2 and 3 |

- 23 Ethene reacts with cold alkaline manganate(VII) ions to form diols, as shown below.



But-1-ene and but-2-ene react with cold alkaline manganate(VII) ions in a similar manner.

Which of the following statements are correct?

- 1: One of the diol formed contains two primary alcohol groups.
- 2: One of the diol formed contains two secondary alcohol groups.
- 3: One of the diol formed contains a primary and a secondary alcohol group.
- 4: Both alkenes exhibit cis-trans isomerism.

- | | |
|-----------------------|-----------------------|
| A 1 and 3 only | B 1 and 4 only |
| C 2 and 3 only | D 2 and 4 only |

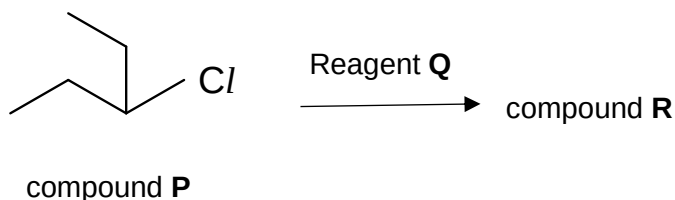
- 24 0.006 mol of alcohol **M** is burnt in excess oxygen and the gases formed are passed through an excess of calcium hydroxide. 3.0 g of calcium carbonate is produced.

M turns hot acidified potassium dichromate(VI) from orange to green.

What could be the identity of **M**?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- C $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$

- 25 Compound **P** reacts with reagent **Q** to form a hydrocarbon, compound **R**.



Which of the following is correct?

- 1 This is an elimination reaction.
- 2 Reagent **Q** is aqueous NaOH.
- 3 Compound **P** is 1-chloro-1-ethylpropane.
- 4 Compound **R** has the empirical formula of CH_2 .

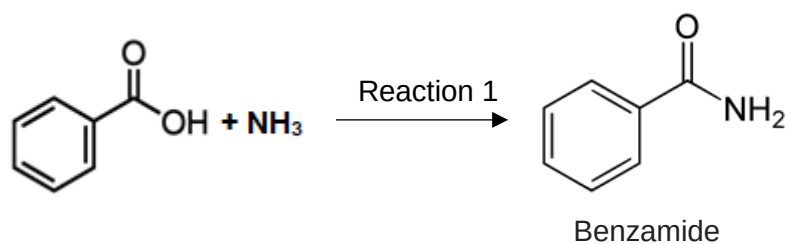
- A 1 and 4 only B 2 and 3 only C 1, 3 and 4 only D All of the above

- 26 1 mole of compound **Q** gives the same volume of gas when reacted with excess sodium metal and sodium carbonate.

Which of the following compound would be compound **Q**?

- A $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- B $\text{CH}_3\text{CH}(\text{OH})\text{CHO}$
- C $\text{HOOCCH}_2\text{CH}_2\text{COOH}$
- D $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_3$

- 27 Benzamide can be synthesise from benzoic acid.

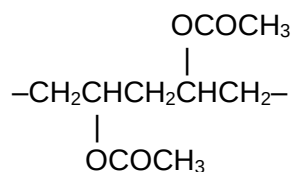


Which of the following is correct?

	Reagent and condition for Reaction 1	Products of Base hydrolysis of Benzamide
A	Concentrated H_2SO_4 heat	
B	Concentrated H_2SO_4 heat	
C	Dicyclohexylcarbodiimide	
D	Dicyclohexylcarbodiimide	

- 28 Glue can be made from the polymer, polyvinylacetate.

A portion of the chain of the polymer is shown.

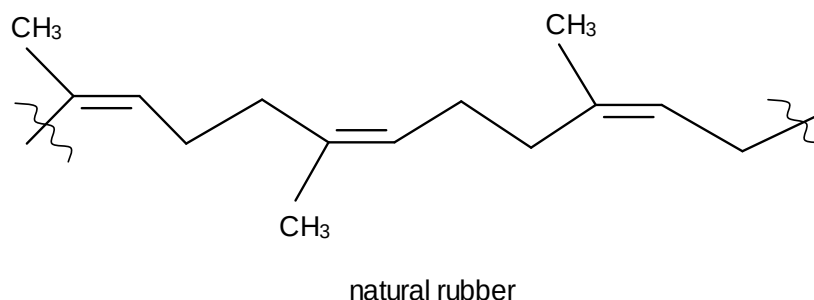
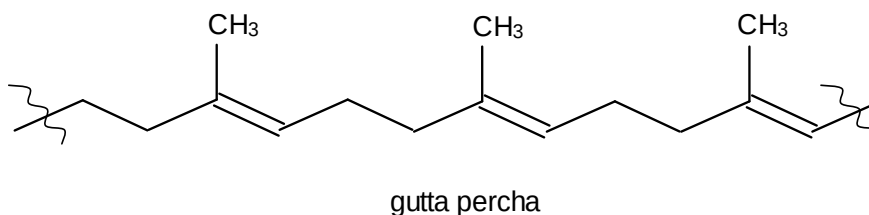


A molecule of this polymer contains 40 carbon atoms.

How many monomer units are incorporated in one molecule of this polymer?

- A** 4 **B** 5 **C** 8 **D** 10

- 29 Gutta percha is a highly crystalline polymer, while natural rubber is a noncrystalline polymer. A section of the two polymers is shown below.



Which statement is true?

- A Gutta percha is more rigid than natural rubber.
 - B Gutta percha and natural rubber have different monomers.
 - C The polymer chains of gutta percha can be packed less closely than natural rubber.
 - D The melting point of gutta percha is lower than natural rubber.
- 30 The gecko can climb up a smooth glass window. The gecko has millions of microscopic hairs on its toes and each hair has thousands of pads at its tips. The result is that the molecules in the pads are extremely close to the glass surface on which the gecko is climbing.
- Which of the following explains gecko's ability to climb on the glass surface?
- A It is due to the ionic interactions between the hairs on the gecko's toes and the surface of the glass surface.
 - B It is due to the cumulative instantaneous dipole – induced dipole interactions between the toes of the gecko and the glass surface.
 - C It is due to the extensive coordinate bonds formed between the toes of the gecko and the glass surface.
 - D It is due to the suction force formed due to the vacuum formed between the toes of the gecko and the glass surface.